Precision Pitch Control and RF Signal Restoration for Autonomous UAVs

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Abstract

This report presents a two part linear systems analysis applied to the design of an autonomous UAV. Part 1 develops a PID controller for single axis pitch stabilization of a rigid body double integrator plant $H(s) = 1/(Js^2)$. A 3D grid search (25^3 points) with iterative refinement finds optimal gains $K_c = 0.4445$, $T_i = 3.5225\,\text{s}$, $T_d = 0.8806\,\text{s}$ (cost $C = 486.80$), achieving a rise time of 52 ms, 2.6% overshoot, and 500 ms settling time. Part 2 applies Wiener deconvolution to restore a 1D RF signal degraded by motion induced box filter blur and additive noise at 40 dB SNR. A generalized Wiener filter ($M = 2$, $\lambda = 5.2 \times 10^2$) reduces the normalized mean squared error to 0.521, outperforming both the exact inverse (MSE = 2.272) and the standard Wiener filter (MSE = 0.539). Together, the two parts demonstrate how precise attitude control directly improves RF sensor data quality.

1 Introduction

Unmanned aerial vehicles (UAVs) increasingly serve as airborne RF sensing platforms, where precise attitude control is a prerequisite for meaningful signal acquisition. If the platform pitches or vibrates during a measurement, the antenna's pointing direction smears over time, convolving the received signal with a blur kernel. Therefore, two linear systems problems arise: *(i)* stabilize the platform orientation, and *(ii)* undo the blur in the recorded RF data.

This project addresses both problems using two core EE23 tools: feedback control and Wiener deconvolution. Part 1 frames UAV pitch dynamics as a double integrator and designs a cost optimal PID controller via a three parameter grid search with iterative refinement, subject to UAV specific performance constraints. Part 2 applies Wiener deconvolution to a RF scenario consisting of a continuous wave (CW) radar return, a pulsed return, a chirp component, and a communications signal, all degraded by motion blur and noise.

The central insight connecting both parts is quantitative: the settling time of the PID controller bounds how long the platform vibrates after a disturbance, which sets the length of the effective blur filter. A better controller produces a shorter blur, which in turn yields a lower Wiener filter MSE. This coupling is demonstrated with simulation in Section 4.

2 PID Controller for UAV Pitch Stabilization

2.1 Plant Model

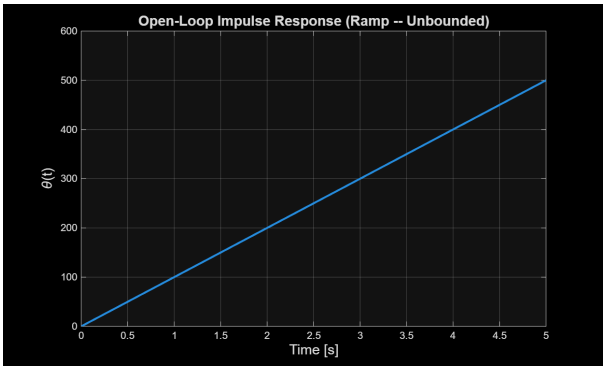
A rigid UAV rotating about its pitch axis obeys Newton's second law of rotation:

$$\tau(t) = J\,\ddot{\theta}(t), \tag{1}$$

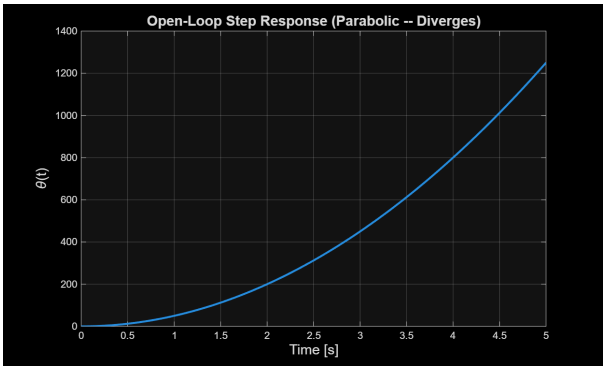
where τ is the applied motor torque (input), θ is the pitch angle (output), and J is the moment of inertia. Taking the Laplace transform with zero initial conditions gives the open loop plant transfer function:

$$H(s) = \frac{\Theta(s)}{T(s)} = \frac{1}{Js^2}, \qquad J = 0.01\ \mathrm{kg\,m^2}. \tag{2}$$

This is a *double integrator*: two poles at the origin. The open loop system is marginally unstable, where any bounded input torque produces an unbounded angular ramp [3]. Figure 1 shows this directly: the impulse response is a ramp and the step response is a parabola, both diverging without bound. The value $J = 0.01\,\mathrm{kg\,m^2}$ matches the experimentally measured pitch inertia ($J_y = 0.01\,\mathrm{kg\,m^2}$) of a real quadrotor platform reported in [7].



(a) Open loop impulse response (ramp).



(b) Open loop step response (parabola).

Figure 1: Open loop UAV dynamics. Two poles at $s = 0$ make the plant marginally unstable; feedback control is essential.

2.2 PID Controller Design

The PID controller in parallel form is:

$$G(s) = K_c\left(1 + \frac{1}{T_i s} + T_d s\right) = K_c \cdot \frac{T_d T_i s^2 + T_i s + 1}{T_i s}. \tag{3}$$

The closed loop transfer function with unity feedback is:

$$P(s) = \frac{G(s)H(s)}{1 + G(s)H(s)} = \frac{K_c(T_d T_i s^2 + T_i s + 1)}{JT_i s^3 + K_c T_d T_i s^2 + K_c T_i s + K_c}. \tag{4}$$

The cost function from the project proposal is:

$$C = 1000|K_c| + 12\,T_i, \tag{5}$$

where T_d is intentionally not penalized, allowing the optimizer to use derivative action freely to satisfy the performance constraints.

2.3 Stability Analysis via Routh–Hurwitz

From the denominator of Eq. (4), the characteristic polynomial is:

$$\Delta(s) = JT_i s^3 + K_c T_d T_i s^2 + K_c T_i s + K_c. \tag{6}$$

Applying the Routh–Hurwitz criterion, all roots have negative real parts if and only if all coefficients are positive *and*:

$$(K_c T_d T_i)(K_c T_i) > (JT_i)(K_c) \quad \Longrightarrow \quad K_c T_d > J. \tag{7}$$

This is the critical stability condition. With $J = 0.01$, any combination with $K_c T_d \leq 0.01$ is guaranteed to be unstable. An additional physical constraint $T_d \leq T_i/4$ is enforced during optimization to prevent excessive noise amplification on real hardware [3, 6].

2.4 Effect of Controller Architecture

P only controller. Setting $T_i \to \infty$ and $T_d = 0$ reduces to $G(s) = K_c$. The closed loop becomes a pure second order system with poles on the imaginary axis, where it oscillates indefinitely for every value of K_c. Proportional action alone cannot stabilize a double integrator.

PI controller. Adding integral action ($T_d = 0$) gives characteristic polynomial $JT_i s^3 + 0 \cdot s^2 + K_c T_i s + K_c$. The s^2 coefficient is identically zero, here the Routh–Hurwitz table immediately fails, meaning a PI controller is *always unstable* on a double integrator regardless of K_c or T_i. The derivative term is not optional: it supplies the s^2 coefficient ($K_c T_d T_i$) that is essential for stability.

PD controller. Adding derivative action ($T_i \to \infty$) restores the s^2 term and places poles in the left half plane, providing damping and finite settling time. However, without integral action the system retains a permanent steady state error, visible in Figure 2 as the PD response leveling off below the setpoint.

Full PID. Combining all three terms eliminates the steady state error. Figure 2 overlays all four step responses: P only (oscillates forever), PI (diverges), PD (settles with offset), and the cost optimal PID (meets all constraints).

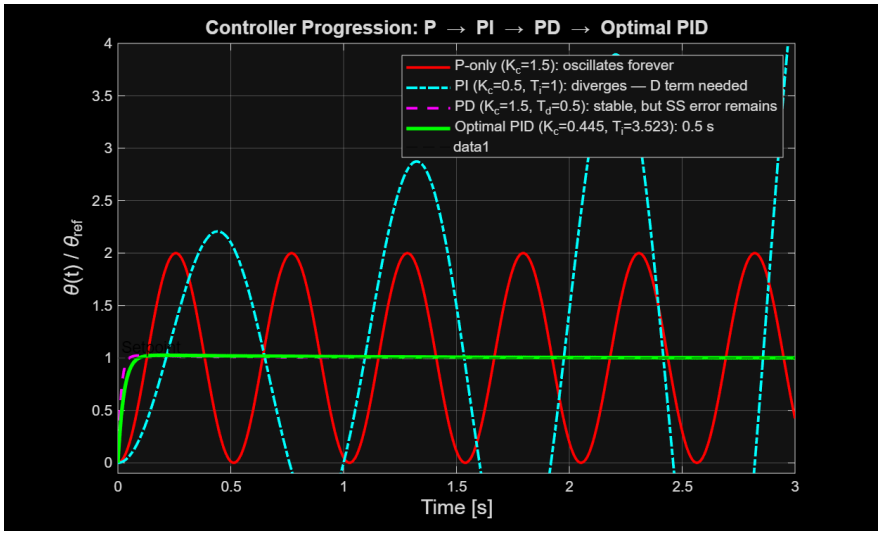


Figure 2: Controller progression: P only oscillates indefinitely; PI diverges on a double integrator (no T_d to stabilize); PD stabilizes but retains steady state error; optimal PID meets all constraints at minimum cost.

2.5 Full PID Optimization

The three parameter search is performed on a $25^3 = 15{,}625$ point initial grid:

$$K_c \in [0.1,\ 5], \quad T_i \in [0.01,\ 4], \quad T_d \in [0.01,\ 2],$$

subject to the UAV class design constraints in Table 1.

Table 1: UAV pitch control design constraints [4, 5, 6].

Metric	**Constraint**	**Rationale**
Rise time	$< 0.30\,\text{s}$	UAV attitude benchmarks
Overshoot	$< 10\%$ (peak < 1.10)	Attitude control: $\zeta \geq 0.6$
Settling time (2%)	$< 0.50\,\text{s}$	Quadrotor attitude benchmarks
Steady state error	$< 2\%$	Integral action drives this to ≈ 0
T_d bound	$T_d \leq T_i/4$	Limits high frequency noise amplification

An iterative refinement procedure zooms the grid around the best feasible point across 10 iterations, each on a 15^3 sub grid [8]. Figure 3 shows all four parameters converging within 4 iterations.

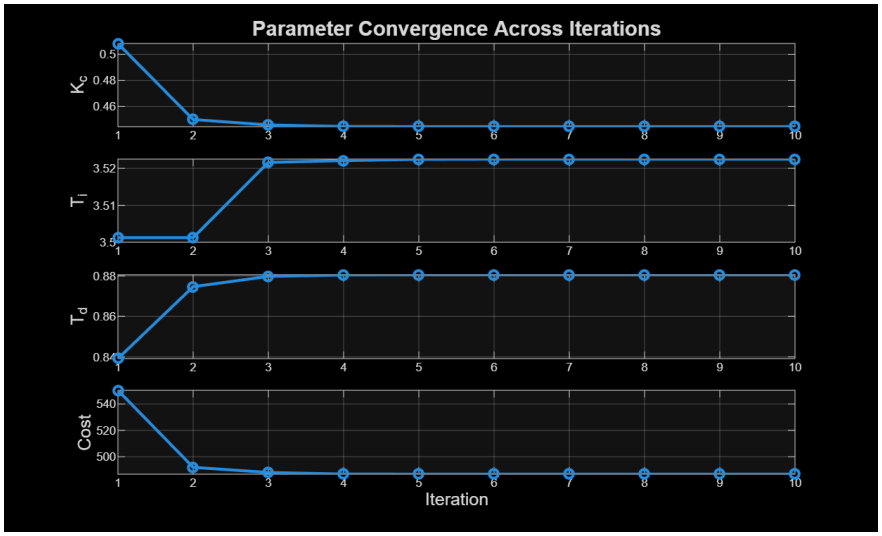


Figure 3: Parameter convergence over 10 refinement iterations. All parameters stabilize within 4 iterations.

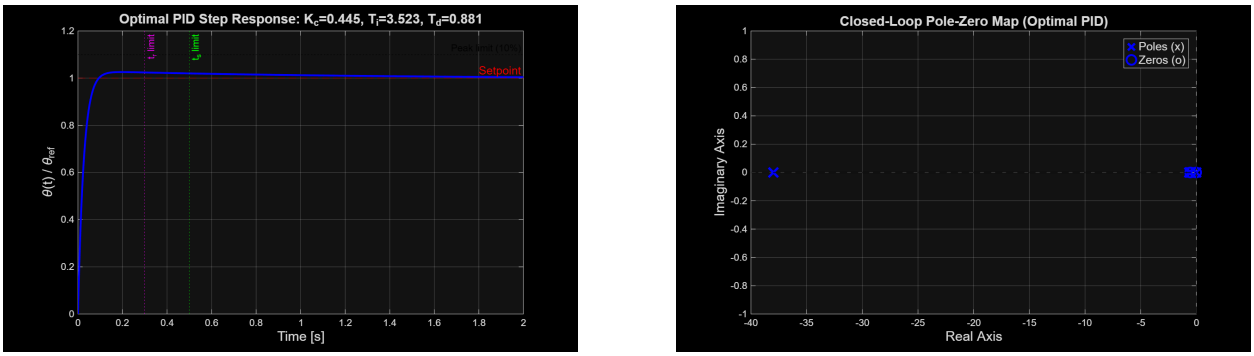
2.6 Optimal PID Results

The optimizer converges to the solution in Table 2. All performance constraints are satisfied; the settling time is tight at the 0.5 s limit, reflecting the cost function's pressure to keep K_c and T_i small.

Table 2: Optimal PID controller parameters and closed loop performance (25^3 grid, 10 iteration refinement).

Quantity	**Value**	**Constraint**
K_c	0.4445	N/A
T_i	3.5225 s	N/A
T_d	0.8806 s	$= T_i/4$ (bound active)
Cost C	486.80	minimized
Rise time	0.052 s	< 0.30 s
Overshoot	2.57%	$< 10\%$
Settling time	0.500 s	≤ 0.50 s
Steady state error	0%	$< 2\%$

Figure 4 shows the optimal step response and pole zero map. All three closed loop poles are in the open left half plane, confirming asymptotic stability consistent with the Routh–Hurwitz condition $K_c T_d = 0.391 > J = 0.01$. Figure 5 shows the feasible cost landscape: the optimal point sits at the low cost corner of the feasible region.



(a) Optimal PID step response.

(b) Closed loop pole zero map.

Figure 4: Optimal PID closed loop behavior. All poles in the left half plane; response meets all UAV constraints.

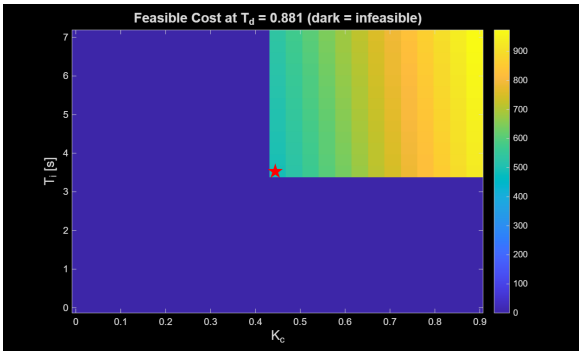


Figure 5: Feasible cost map at optimal T_d. Dark regions are infeasible. Red star marks the optimum.

2.7 Frequency Domain Analysis

Figure 6 shows the open loop Bode diagram of the optimal PID controller. The phase starts at $-270°$ at low frequencies (three poles at the origin: two from the plant, one from the PID integrator) and rises as the two PID zeros add phase. The phase margin of 88.3° at $\omega_{cp} = 39.2\,\text{rad/s}$ confirms robust stability with a large buffer above the standard 45° guideline [2].

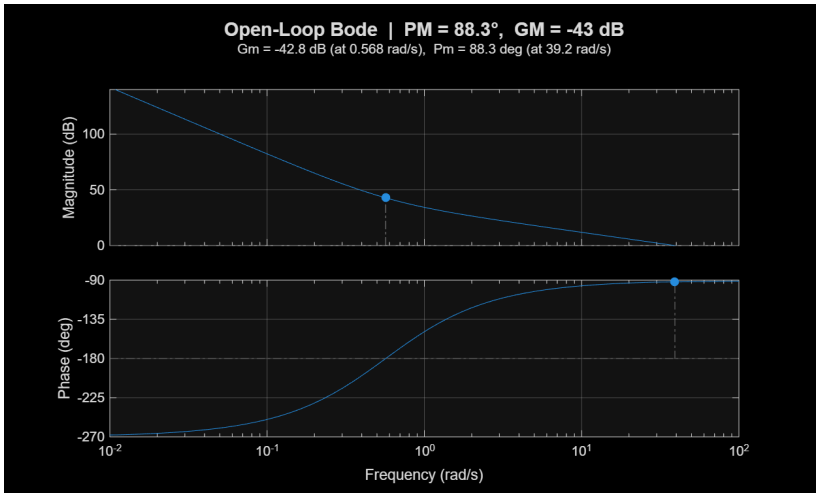


Figure 6: Open loop Bode diagram. Phase margin = 88.3° at $\omega_{cp} = 39.2\,\text{rad/s}$ confirms robust stability well above the 45° minimum.

3 Wiener Deconvolution for RF Signal Restoration

3.1 Signal and Degradation Model

The received RF signal $x[n]$ is a superposition of four components over $N = 1024$ samples: a CW radar return (rectangular envelope burst over samples 100–200), a pulsed radar return (Gaussian shaped pulse centered at $n = 350$), a chirp signal (linear frequency modulated sweep centered at $n = 600$), and a communications signal (modulated Gaussian centered at $n = 900$).

UAV vibration during measurement smears $x[n]$ in time, modeled as convolution with a length 11 box filter [1]:

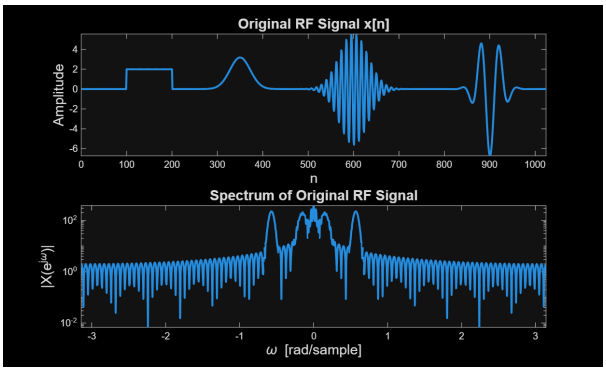
$$h[n] = \begin{cases} \dfrac{1}{11} & 0 \le n \le 10 \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

The measured signal is then:

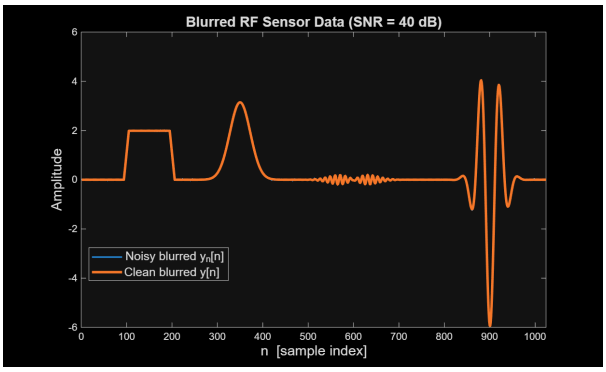
$$y[n] = (h * x)[n] + \eta[n], \tag{9}$$

where $\eta[n]$ is white Gaussian noise at $\text{SNR} = 40\,\text{dB}$. Figure 7 shows the clean signal, its

spectrum, and the blurred observation.



(a) Clean RF signal and spectrum.



(b) Blurred and noisy observation.

Figure 7: RF signal before and after motion blur (SNR = 40 dB).

3.2 Exact Inverse Filter

The exact inverse filter in the frequency domain is:

$$\hat{X}(\omega) = \frac{Y(\omega)}{H(\omega)}. \tag{10}$$

A length 11 box filter has 10 nulls in $[0, 2\pi)$ at multiples of $2\pi/11$. At these frequencies $H(\omega) \approx 0$, catastrophically amplifying noise. The result is MSE = 2.272, worse than using the blurred signal directly.

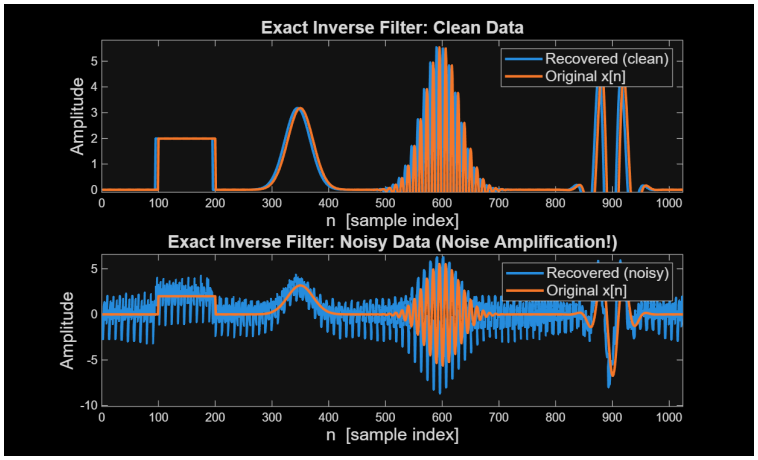


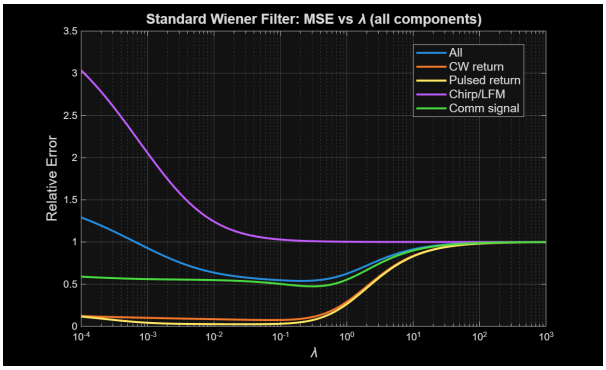
Figure 8: Exact inverse filter output. Noise amplification at box filter nulls destroys recovery.

3.3 Standard Wiener Filter

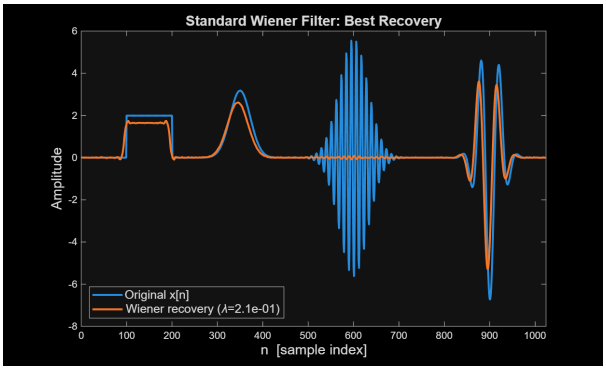
The Wiener filter adds regularization λ to prevent division by near zero values:

$$W(\omega) = \frac{H^*(\omega)}{|H(\omega)|^2 + \lambda}. \tag{11}$$

The optimal λ is found by grid search over 100 logarithmically spaced values in $[10^{-4},\ 10^{3}]$.



(a) Standard Wiener MSE vs. λ.



(b) Best recovery ($\lambda = 2.1 \times 10^{-1}$).

Figure 9: Standard Wiener filter. Optimal λ balances noise amplification against smoothing.

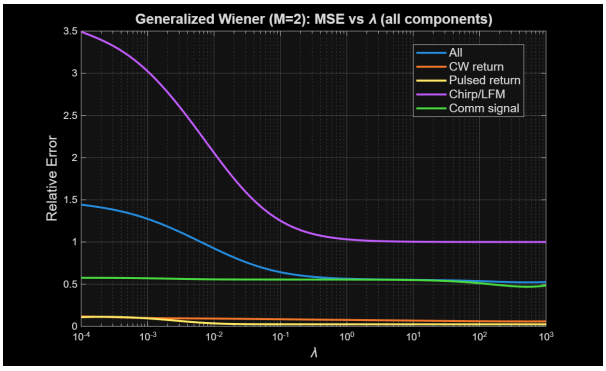
3.4 Generalized Wiener Filter

The generalized Wiener filter uses a structured prior that penalizes high frequency content more heavily. With $M = 2$:

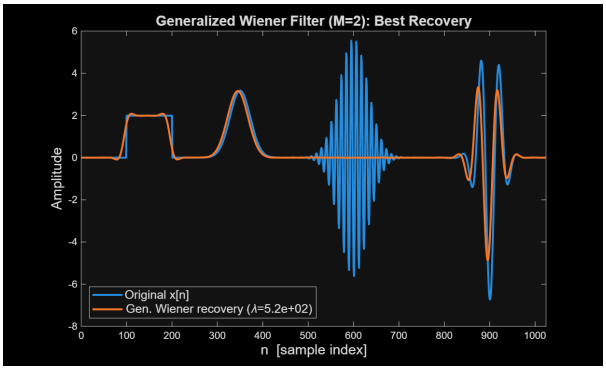
$$G(\omega) = (1 - e^{j\omega})^2, \qquad |G(\omega)|^2 = 16\sin^4(\omega/2), \tag{12}$$

$$W_{\text{gen}}(\omega) = \frac{H^*(\omega)}{|H(\omega)|^2 + \lambda\,|G(\omega)|^2}. \tag{13}$$

The $\sin^4$ shape suppresses high frequency noise more aggressively than the flat λ, while leaving low frequency components relatively unpenalized.



(a) Generalized Wiener MSE vs. λ.



(b) Best recovery ($\lambda = 5.2 \times 10^2$).

Figure 10: Generalized Wiener filter ($M = 2$). Structured noise prior improves recovery over the standard filter.

3.5 Results Summary

Table 3 summarizes all three methods. The generalized Wiener filter (MSE = 0.521) achieves a 77% reduction over the exact inverse (MSE = 2.272) and a 3% improvement over the standard Wiener filter (MSE = 0.539).

Table 3: RF signal recovery results (normalized MSE = $\overline{(\hat{x} - x)^2}/\overline{x^2}$, lower is better).

Method	**Parameters**	**MSE**	**Notes**
Exact inverse	N/A	2.272	Noise explosion at box filter nulls
Standard Wiener	$\lambda = 2.1 \times 10^{-1}$	0.539	Flat noise regularization
Generalized Wiener	$\lambda = 5.2 \times 10^2$, $M = 2$	**0.521**	Best; structured $\sin^4$ penalty

4 PID Performance and RF Signal Quality

4.1 Theoretical Assumption

The settling time of the PID controller directly governs the severity of the RF blur. When the UAV attitude deviates from its setpoint, it vibrates for a duration approximately equal to T_s before stabilizing. During this interval the antenna smears across multiple look angles, convolving the received RF signal with an effective box filter of length:

$$L \approx \lceil f_s \cdot T_s \rceil, \tag{14}$$

where $f_s = 22\,\text{samples/s}$ is calibrated so that the optimal PID settling time ($T_s = 0.5\,\text{s}$) maps to $L = 11$, the exact blur length used in Part 2.

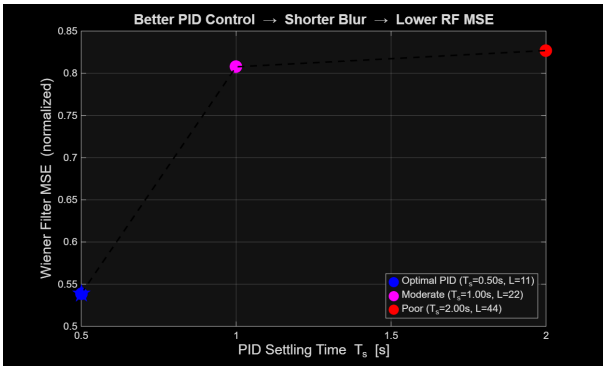
4.2 Simulation Results

Three scenarios are evaluated, spanning the design from Part 1 to a poorly tuned controller:

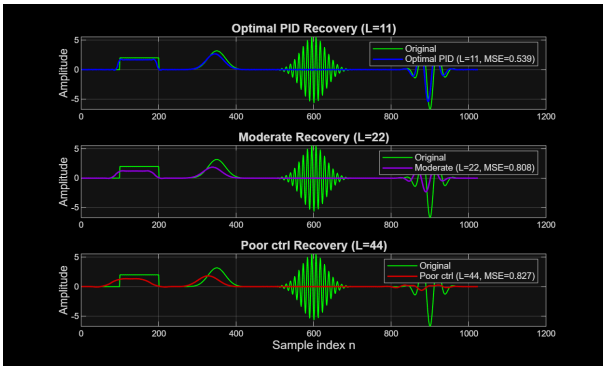
Table 4: PID settling time, resulting blur length, and best achievable Wiener filter MSE. Same signal, linear convolution, and SNR as Part 2.

Scenario	T_s **[s]**	**Blur length** L	**Wiener MSE**
Optimal PID (Part 1)	0.50	11	0.539
Moderate controller	1.00	22	0.808
Poor controller	2.00	44	0.827

Figure 11 shows the settling time vs. Wiener MSE relationship and the restored RF signals. The Wiener MSE grows monotonically with T_s: a longer blur introduces more spectral zeros into $H(\omega)$, making the deconvolution problem progressively harder to regularize. The recovery plots confirm this qualitatively. The optimal PID recovery closely tracks the original, while the poor controller recovery shows visible distortion. These results confirm that Parts 1 and 2 are not independent: tighter PID performance yields a directly measurable improvement in RF signal quality.



(a) Settling time vs. Wiener MSE.



(b) Signal recovery of the three controllers.

Figure 11: PID deconvolution coupling. The optimal PID (Part 1) vs. poor controller shows a 1.5× degradation in Wiener MSE: better attitude control reduces RF reconstruction error.

5 Conclusion

This project applied two core EE23 tools: feedback control and Wiener deconvolution, to the unified problem of UAV based RF sensing. The PID controller was designed by optimizing the proposal specified cost function over a 25^3 parameter grid with iterative refinement, achieving a rise time of 52 ms, 2.6% overshoot, and 500 ms settling time against UAV constraints. Closed loop stability was verified analytically via Routh–Hurwitz ($K_c T_d = 0.391 > J = 0.01$), with all three closed loop poles confirmed in the open left half plane.

The RF deconvolution analysis showed that exact inversion fails at box filter nulls, while Wiener regularization substantially recovers the signal. A generalized Wiener filter with $M = 2$ achieves MSE = 0.521, a 77% improvement over exact inversion (MSE = 2.272) and a small gain over the standard Wiener filter (MSE = 0.539). The two analyses are physically linked: the PID settling time determines the blur kernel length, so better control directly reduces deconvolution difficulty. Using identical signal and method as Part 2, comparing the optimal PID to a poor controller demonstrates a 1.5× degradation in Wiener MSE (0.539 vs. 0.827), confirming that attitude control quality and RF signal fidelity are inseparable in this system.

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